mHCI

Lecture 8:
Mobile Communication
Lab. vs. Field Studies

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Goals of this lecture

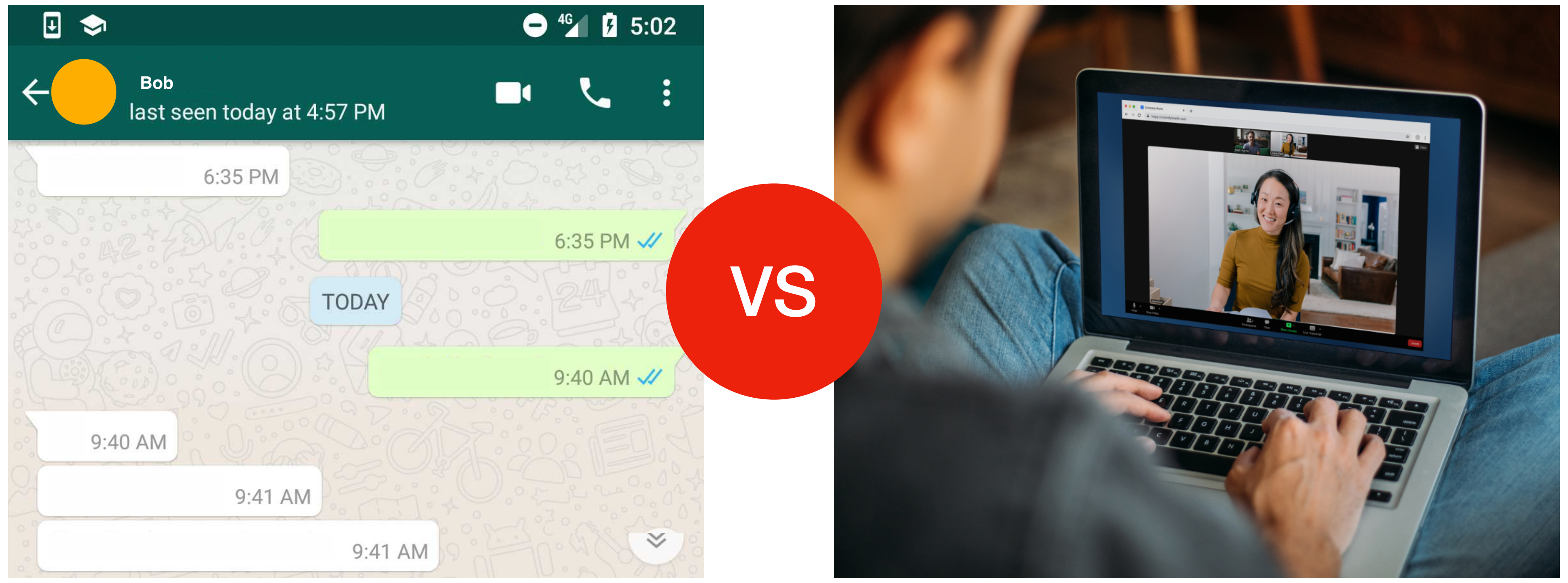
- Key concepts about mobile communication
  - Rich media theory
  - Social presence / co-presence
  - (Selective) self-presentation
- Lab vs. field experiments
- One-off vs. Longitudinal studies
- Control vs. Realism in experimental settings
- Controlling the Context of Use in experimental settings

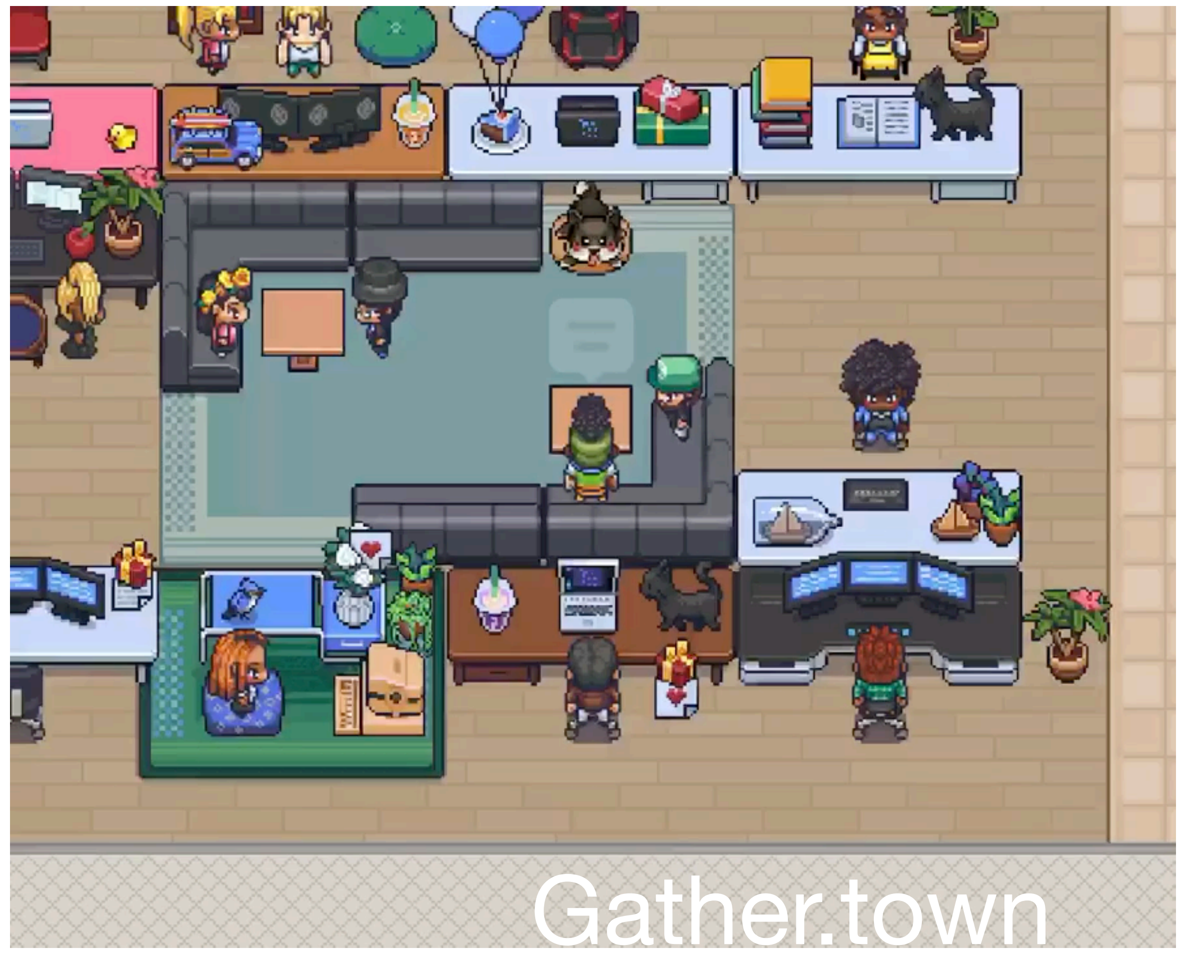
“Together but not together”: Evaluating Typing Indicators for Interaction-Rich Communication

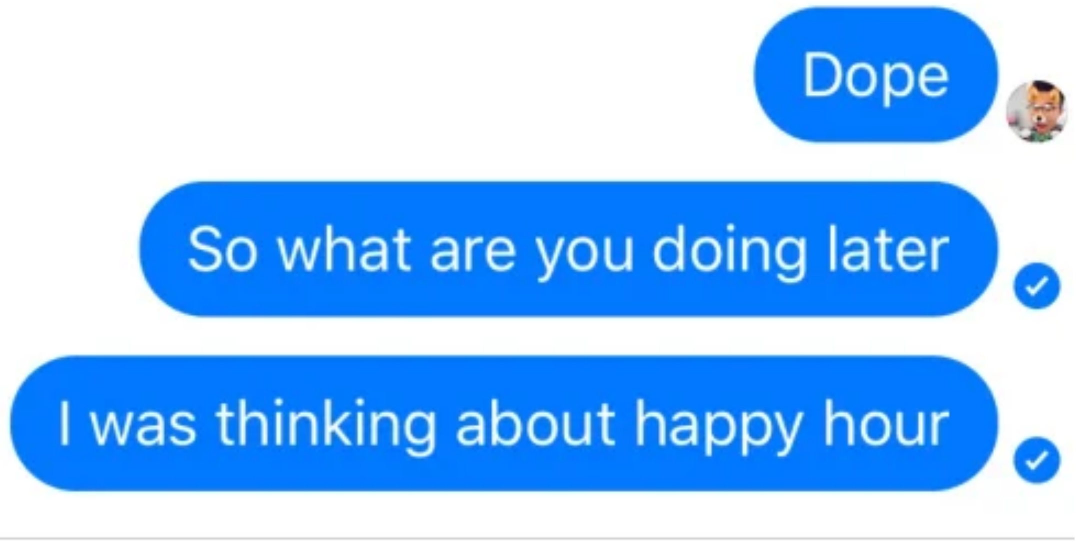
Zainab Iftikhar, Yumeng Ma, Jef Huang
Published in CHI 2023
https://doi.org/10.1145/3544548.3581248

Motivation and Research Questions

- Media Richness Theory:
    - Richer, personal communication media are generally more effective for communicating complex messages in contrast with leaner, less rich media
- Social presence / co-presence:
    - The sense of "being together" in the same space—digital communication platforms can vary in how much they can support a sense of "being with another" compared to face-to-face interactions
- Selective self-presentation:
    - How people leverage the limitations communication media to "present themselves" to shape how others view them







https://www.gather.town/

https://zoom.us/

https://mashable.com/article/how-to-read-facebook-messenger-messages-no-read-receipts

Motivation and Research Questions

Richer media can potentially increase **social presence** in online communication. However, at the same time, it might limit possibilities for **selective self-presentation**.

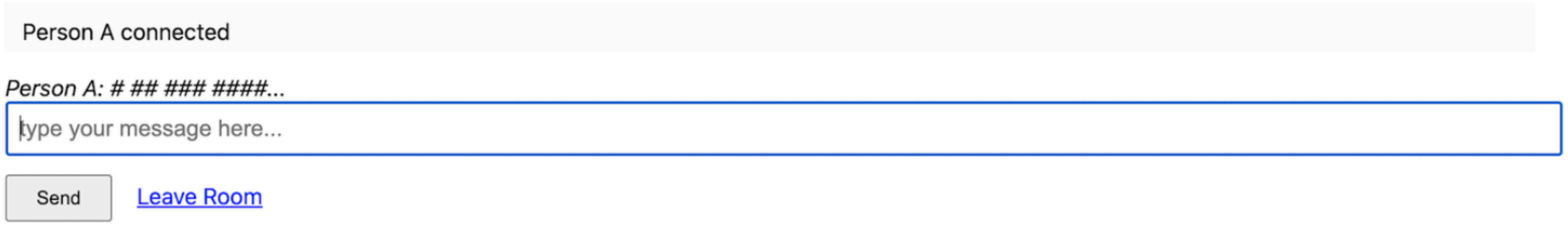
In the context of text-based communication during collaborative or problem-solving tasks, **does the *richness* of a medium have an impact on task performance and overall user experience?**

What kind of study would be suitable to answer this question?

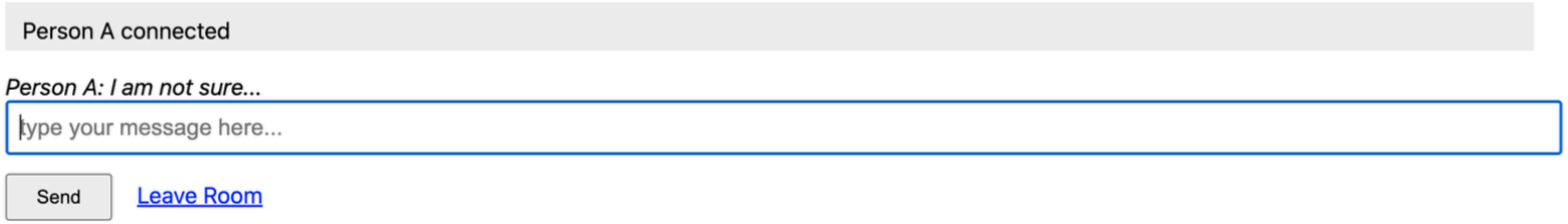
What do you think about this?

User Study: Remote experiment

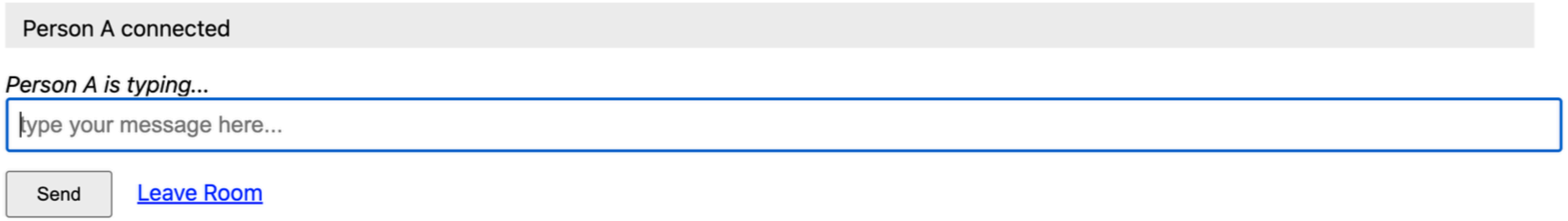
"We conducted a remote study to allow participants to experience the typing indicators without the constrictions of a lab setting”



Masked typing



Live typing



Is typing

No typing indicator

User Study: Remote experiment

- What are the hypotheses?
- What's the task?
- What are the independent variables?
- What are the dependent variables?
- How are participants allocated to independent variables? Is it a within-subjects design or between subjects design?

User Study: Remote experiment

Task: chat with another participant to make a decision about Problem 1/2/3/4, with a time limit of 8 minutes.

Participants: 24 total. 3 sets of 8 individuals

Independent variables:

Primary: Typing indicator (Room), 4 levels

Secondary: Problem to discuss during each trial, 4 levels

Dependent variables (measures):

NASA TLX

helpfulness and effectiveness of each indicator

# of typed words and sent messages

Data collection methods: NASA TLX questionnaire, end-of-trial and end-of-experiment questionnaire + semi-structured interview, typing logs

	Room A	Room B	Room C	Room D
Problem 1	(P6,P7)	(P1,P5)	(P2,P4)	(P3,P8)
Problem 2	(P1,P8)	(P4,P7)	(P3,P5)	(P2,P6)
Problem 3	(P3,P4)	(P2,P8)	(P1,P6)	(P5,P7)
Problem 4	(P2,P5)	(P3,P6)	(P7,P8)	(P1,P4)

Each room has a different typing indicator condition

Tasks differed in their survival scenario and the list of items that were to be ranked by the participants. We selected the following four parallel versions of the survival tasks for balanced complexity:

- Desert Survival Task with validation provided by the Chief of the Desert Branch [36]
- NASA Moon Survival Task with validation provided by NASA experts [26]
- Lost at Sea Task validated by the US Coast Guard [45]
- Plane Crash Task validated by the US Army [33]

NASA Task Load Index (TLX)

Hart and Staveland's NASA Task Load Index (TLX) method asseses work load on five 7-points scales. Increments of high, medium and low estimates for each point result in 21 gradations on the scales.

Name	Task	Date

Mental Demand How mentally demanding was the task?

Very Low Very High

Physical Demand How physically demanding was the task?

Very Low Very High

Temporal Demand How hurried or rushed was the pace of the task?

Very Low Very High

Performance How successful were you in accomplishing what you were asked to do?

Perfect Failure

Effort How hard did you have to work to accomplish your level of performance?

Very Low Very High

Frustration How insecure, discouraged, irritated, stressed, and annoyed were you?

Very Low Very High

User Study: Quantitative Results

Any interesting insights coming from the *quantitative* results?

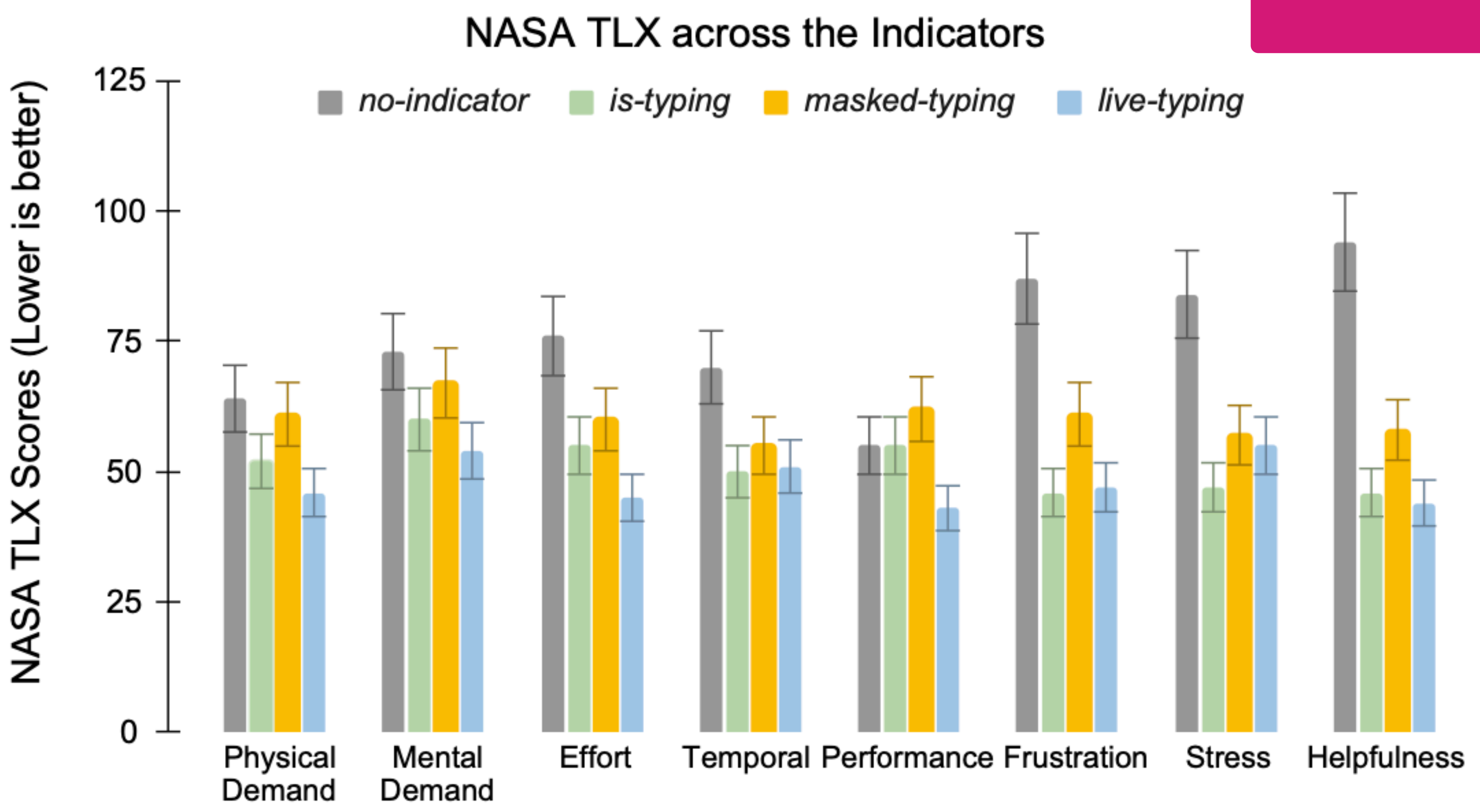


Figure 6: NASA Task Load Index assessment shows that *live-typing* significantly causes less *Frustration* and *Stress*, has lower *Temporal*, *Physical*, and *Mental Demand*, higher perceived *Performance*, and has less perceived *Effort*

	No Indicator	Is Typing	Masked Typing	Live Typing
Total	210.00	248.00	190.00	246.00
Mean	17.50	20.71	15.85	20.50
SD	7.42	6.72	5.87	8.05

(No statistically significant difference across typing indicators)

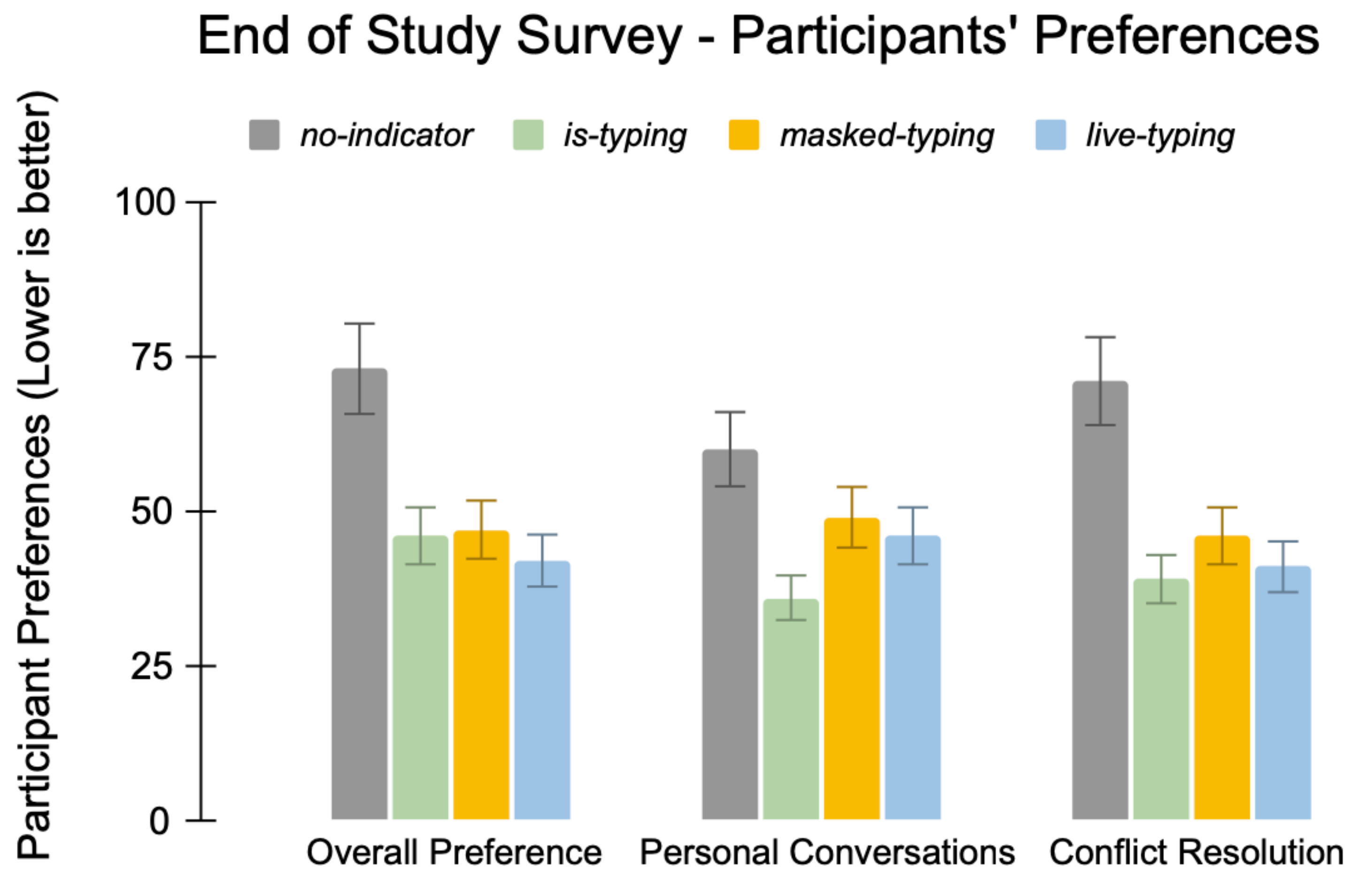


Figure 7: End-of-study survey shows that participants overall preferred *live-typing* however for personal lives and resolving conflict, most participants preferred the *is-typing* interface. None of the participants opted for the baseline (*no-indicator*).

User Study: Qualitative Results

Any interesting insights coming from the *qualitative* results?

5.2.1 Richer texting platforms are less frustrating and cognitively demanding.

Most participants valued the heightened presence and immediate feedback in the new indicators. For instance, P11 reported that they *"just needed to know someone is working with me on the task"* whereas P18 valued the "involving nature" of the indicator. These findings

5.2.2 Richer texting platforms can be validating encouraging active listening.

P14 echoed this and felt that *live-typing* helped them understand the other person better: *"Seeing what they are typing and erasing in real-time helped me pay more attention like someone is speaking their mind on the go."*

5.2.3 Richer texting platforms limit selective self-presentation.

Five participants (20%) felt uncomfortable and found that either one or both transparent interfaces limited them in their discussions. For instance, P11 said that *live-typing* was *"like walking on eggshells"* and P21 found that they were *"formulating the perfect response in their mind before saying (typing) anything."* When informed that the

Findings and Conclusions

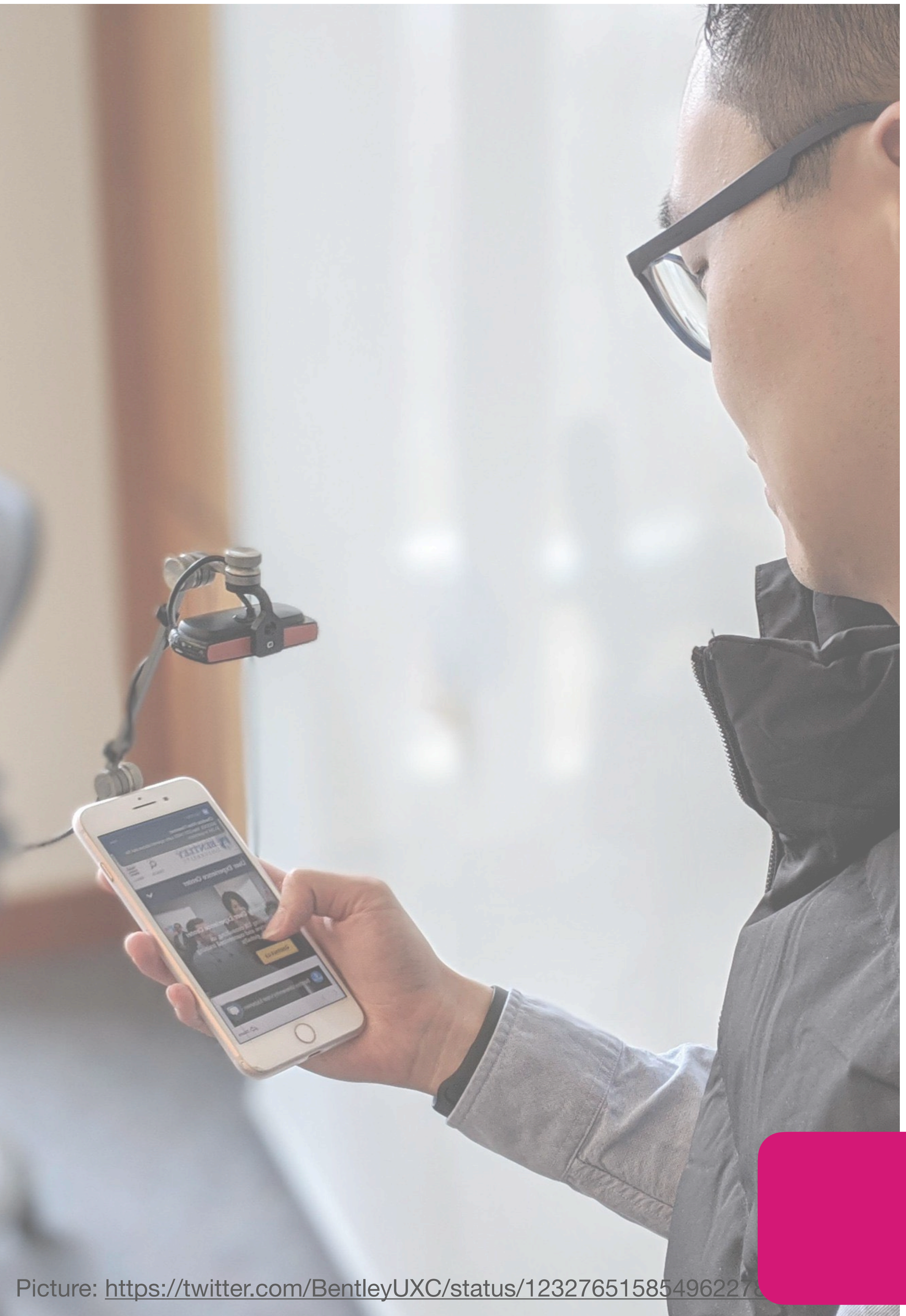
- According to the quantitative measures, masked-typing helped users reach more agreements on the study task
- However, the qualitative data and analysis suggests that live-typing is perceived as more communicative, helpful, and effective for their task-based interaction.

- Live-typing increased users' perceived social presence
- However, participants were apprehensive about the applicability of the indicator in their daily lives as it limited their selective self-presentation

Discussion time

- Strengths: they did a good job in pointing out contradictory findings coming from different measures (e.g., that masked typing performed best for the task, but participant liked live-typing more… though not for everyday communication).
- Limitations: Why NASA-TLX if the goal was to increase social presence?
- Design ideas: Could typing indicators be customizable? Could users choose a different indicator for different conversations? Could they choose it “together" with their conversation partners?

What happens outside the lab?



Lab study	Real Life
Environment is controlled	Environment can be unexpected
Software used in isolation	Software used within a digital ecology
Social isolation	Used in social settings
Tasks with instructions	On-demand tasks
Used within a timeframe	Used when needed

The context of use!

What is being controlled in a lab setting?

Picture: https://twitter.com/BentleyUXC/status/12327651585496227

Recap: Experiments

Experiments are designed for finding **a cause-effect relationship** between a particular system design and the user performance or experience while doing a task with that design.

does this design… **… cause this effect?**

Your Design

smaller / bigger buttons
text input vs. speech input
different menu styles
etc

The user's performance/ experience on a task

time to complete the task
number of errors
Enjoyment/Frustration
etc

Lab experiments control the context of use **to make sure that the effects we observe are caused by our design and not by external factors** (e.g., interruptions, distractions, technical differences across devices, etc.). But will the results represent realistic use?

Lab. vs Field Experiments as a spectrum between control and realism

The more you control the context of use in your experiment, the more likely that your results are caused by your design

The more realistic the context of use in your experiment, the more your results reflect how people actually behave with the technology in real life

We can increase realism to the context of use in the experiment in two ways:

Simulating realistic contexts
(e.g., faking GPS coordinates to simulate location changes)

Not controlling certain types of context
(e.g., asking all participants to use the same device vs. letting them use their own device)

Control vs. Realism of Context of Use in Experiments

	More control (Less realism) ⟷	Less control (More realism)
Physical Context	e.g., all participants use the system in the same place (such as a lab)	e.g., each participant uses the system where they want
Social Context	e.g., participants use the system with assigned experiment partners	e.g., participants use the system with whoever they want
Temporal Context	e.g., participants use the system at specific times in the day	e.g., participants use the system whenever they want
Technical Context	e.g., participants use different versions of the system determined by us	e.g., participants choose what version to use to fit to their needs
	e.g., participants use our pre-set devices	e.g., participants use their own devices
Task Context	Participants are instructed to perform specific tasks	Participants use the system as they see fit